

Middle East Technical University
Department of Computer Engineering
Wireless Systems, Networks and Cybersecurity (WINS) Laboratory



Phase 1: WARM-UP

CENG519 Network Security
2024-2025 Spring
Term Project Report

Prepared by
Mohamed Eldeeb
Student ID: 2643120
mohamed.eldeeb@metu.edu.tr
Computer Engineering
23 March 2025

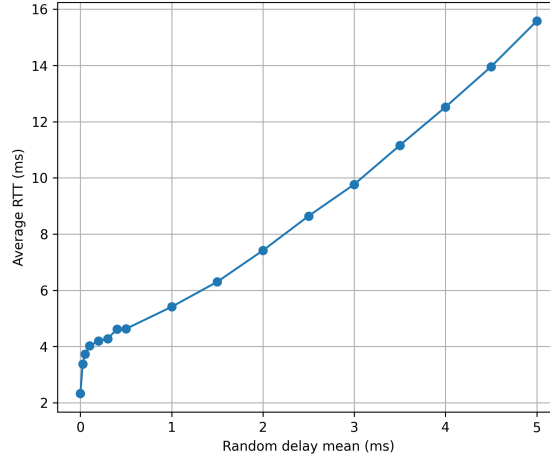


Figure 1 . Average round-trip time for ping packets

1 Introduction

This is the report for the first phase of the term project for Network Security. It explains the relationship between the mean of the random delay added by the processor and the round-trip time (RTT) of the ping packets. Code is available at <https://github.com/mosamadeeb/middlebox/tree/project>.

2 Results and Discussion

2.1 Methodology

The ping command was used to send packets from the secure host to the insecure host. The go-processor was run each time with a different random delay mean, exponentially distributed. A total of 1000 packets were sent at intervals of 10ms for each measurement, and the average RTT was recorded.

2.2 Results

It can be seen from Figure 1 that the relationship between the exponential mean and the average RTT is linear for higher values, specifically higher than 1ms. Switching to a logarithmic scale as in Figure 2 shows the relationship better for smaller values. It can be concluded that RTT is more largely affected by the delay when it is higher, and that a smaller delay would probably model real network delay better.

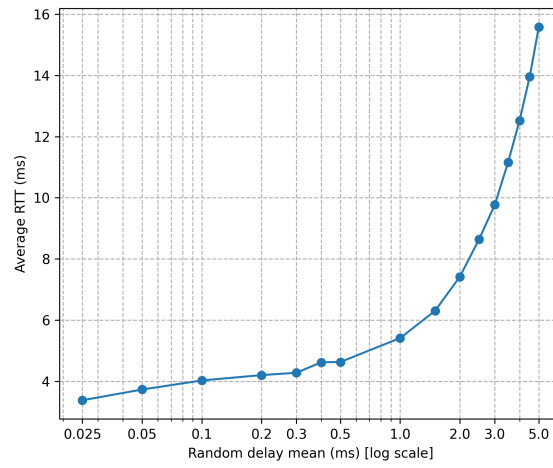


Figure 2 . Average round-trip time for ping packets (logarithmic scale)